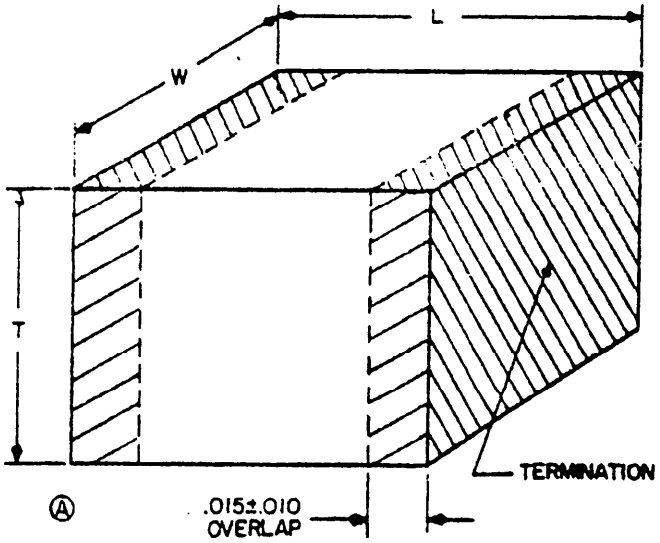
MIL-C-11272/17A(USAF)
5 JULY 1973
SUPERSEDING
MIL-C-11272/17(USAF)
12 January 1973

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, GLASS DIELECTRIC, STYLES CY83 AND CY84

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Military Specification MIL-C-11272.



Ⓐ

Dimensions				
			T	
Style	L	W	Min	Max
CY83	.110 ± .015	.110 ± .015	.060	.100
CY84	.110 ± .020	.110 ± .015	.060	.100

INCHES	MM
.010	.25
.015	.38
.020	.51
.060	1.52
.100	2.54
.110	2.79

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

Ⓐ denotes changes.

FSC 5910

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TABLE I.

Type designation 1/	Capacitance	Rated voltage	Capacitance tolerance available
	pF	Volts, dc	
CY8-D0R1B	0.1	500	B
CY8-D0R2B	0.2		B
CY8-D0R3-	0.3		B, C
CY8-D0R4-	0.4		B, C
CY8-D0R5-	0.5		B, C, D
CY8-D0R6-	0.6		
CY8-D0R7-	0.7		
CY8-D0R8-	0.8		
CY8-D0R9-	0.9		
CY8-D1R0-	1.0		
CY8-D1R1-	1.1		
CY8-D1R2-	1.2		
CY8-D1R3-	1.3		
CY8-D1R4-	1.4		
CY8-D1R5-	1.5		
CY8-D1R6-	1.6		
CY8-D1R7-	1.7		
CY8-D1R8-	1.8		
CY8-D1R9-	1.9		
CY8-D2R0-	2.0		
CY8-D2R1-	2.1		
CY8-D2R2-	2.2		
CY8-D2R4-	2.4		
CY8-D2R7-	2.7		
CY8-D3R0-	3.0		
CY8-D3R3-	3.3		
CY8-D3R6-	3.6		
CY8-D3R9-	3.9		
CY8-D4R3-	4.3		
CY8-D4R7-	4.7		
CY8-D5R1-	5.1		
CY8-D5R6-	5.6		
CY8-D6R2-	6.2		
CY8-D6R8-	6.8		B, C, J, K, M
CY8-D7R5-	7.5		
CY8-D8R2-	8.2		
CY8-D9R1-	9.1		
CY8-D100-	10		F, G, J, K, M
CY8-D110-	11		
CY8-D120-	12		
CY8-D130-	13		
CY8-D150-	15		
CY8-D160-	16		
CY8-D180-	18		
CY8-D200-	20		
CY8-D220-	22		
CY8-D240-	24		
CY8-D270-	27		

See footnote at end of table.

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TABLE I. - Continued

Type designation 1/	Capacitance	Rated voltage	Capacitance tolerance available
	pF	Volts, dc	
CY8-D300-	30	500	F, G, J, K, M
CY8-D330-	33		
CY8-D360-	36		
CY8-D390-	39		
CY8-D430-	43		
CY8-D470-	47		
CY8-D510-	51		
CY8-D560-	56		
CY8-D620-	62		
CY8-D680-	68		
CY8-D750-	75		
CY8-D820-	82		
CY8-D910-	91		
CY8-D101-	100		
CY8-D111-	110	300	
CY8-D121-	120		
CY8-D131-	130		
CY8-D151-	150		
CY8-D161-	160		
CY8-D181-	180		
CY8-D201-	200		
CY8-D221-	220	200	
CY8-D241-	240		
CY8-D271-	270		
CY8-D301-	300		
CY8-D331-	330		
CY8-D361-	360		
CY8-D391-	390		
CY8-D431-	430		
CY8-D471-	470		
CY8-D511-	510	100	
CY8-D561-	560		
CY8-D621-	620		
CY8-D681-	680	50	
CY8-D751-	750		
CY8-D821-	820		
CY8-D911-	910		
CY8-D102-	1000		

1/ Complete type designation will include an additional digit to indicate the style, and an additional letter symbol to indicate the capacitance tolerance, where applicable.

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REQUIREMENTS

Design and construction:
Dimensions and configuration: See figure and table I.
Case type: Multi-layer, unencapsulated, monolithic.
Material - Porcelain.
Terminations:
Style CY83 - Metallized.
(A) Style CY84 - Solder coated, 372° F, 62 Sn, 36Pb, 2 percent Ag.
(A) Inspection conditions: All visual examinations shall be performed using a stereo microscope having a minimum magnification of 20 power.
Capacitance (Cap.) value: See table I.
Cap. tolerance: See table I
Rated voltage: See table I, characteristic D.
Operating temperature range: -55° C to +125° C.

Insulation resistance (IR):
At +25° C: 10^6 megohms, minimum (for 0.1 pF thru 470 pF).
10^5 megohms, minimum (for 510 pF thru 1,000 pF).
At +125° C: 10^5 megohms, minimum (for 0.1 pF thru 470 pF).
10^4 megohms, minimum (for 510 pF thru 1,000 pF).

(A) Quality factor: 10,000. 1 MHz -0 +100 kHz.

Life:
At +25° C and +125° C:
IR - Not less than initial requirement.
(A) Cap. - Change not more than 0.2 percent or 0.2 pF, whichever is greater, from the nominal value.

(A) Temperature coefficient and capacitance drift. See table II.

TABLE II. Temperature coefficient and capacitance drift.

Temperature coefficient	Capacitance drift (-55° to +125° C)
Parts/million/° C 90 ±20	0.1 percent or 0.1 pF, whichever is greater

NOTE: These capacitors are intended for use in RF solid-state circuitry up to and including 25 gigahertz, and cannot be replaced in the field.

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5910-F196-2)

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5. PROBLEM AREAS

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